Identifying Hidden Graphs with Quantum Walks

From BQP-Completeness to an Exponential Separation

arXiv:2605.11228

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What I work on these days

- quantum computing needs **new algorithms**, and **new applications** for the ones we have
- the **Genesis Mission** is aimed at exactly that: using AI for quantum algorithm discovery
- $\Rightarrow$ the **paper** behind this talk, arXiv:2605.11228, was a warm-up for it
  - the ideas are mine, found without AI
  - the **numerics** took non-trivial AI assistance

The problem

We are handed a Hamiltonian H, together with a start state $|\psi\rangle$.

- H **sparse** and **exponentially large**; $|\psi\rangle$ a **computational basis state**
- H and $|\psi\rangle$ define the **signal**:

$$f(t) \; := \; \langle\psi|e^{-iHt}|\psi\rangle$$

- a **pair** of *small* graphs A_0, A_1, and a vertex v, all known to us
- we are **promised** that H acts as A_b on the subspace the evolution explores, for one of $b \in \{0, 1\}$

Decide b from the signal.

The promise in detail: H on the Krylov subspace

- the **Krylov subspace** $\mathcal{K} = \mathrm{span}\{H^k|\psi\rangle\}$: the smallest subspace containing $|\psi\rangle$ that the evolution never leaves
- the **promise** is that $\mathcal{K}$ is **tiny**, and that on it H *is* a graph:

$$|\psi\rangle \leftrightarrow |v\rangle, \qquad H\big|_{\mathcal{K}} = \underbrace{A_b}_{\text{adjacency matrix, small graph}}$$

- the correspondence runs through *every* power, $H^k|\psi\rangle \leftrightarrow A_b^k|v\rangle$
  $\Rightarrow$ the signal transfers:

$$f(t) = \langle\psi|e^{-iHt}|\psi\rangle = \langle v|e^{-iA_bt}|v\rangle$$

A quantum walk: e^{-iA_bt}, on a graph, from a vertex.

An example: one flipped spin on a ring

- N spins on a ring, nearest-neighbour hopping, 2^N dimensions:

$$H \;=\; \sum_j \left(\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+\right) \qquad \text{(2-local)}$$

- start from **one flipped spin**; write $|j\rangle$ for the flip at site j, a computational basis state
- $H|j\rangle \;=\; |j{-}1\rangle + |j{+}1\rangle$: the flip hops, and no others appear
- $\Rightarrow$ $\mathcal{K}$ is the N positions, and $H\big|_{\mathcal{K}} = A(C_N)$

2^N-dimensional local Hamiltonian;
on the Krylov subspace, the N-cycle.

How is H given?

- the algorithm only ever *uses* the signal $f(t) = \langle\psi|e^{-iHt}|\psi\rangle$
- to observe f, it has to *simulate* e^{-iHt}
  $\Rightarrow H$ has to be specified somehow
- **as local terms**: a sum of $O(1)$-local terms, written down explicitly; the example above, and the BQP instance
- **as a sparse black box**: one query returns the nonzero entries of one row; the spired instance
- both are enough to simulate e^{-iHt} efficiently

Local terms, or one row at a time.

A simple quantum algorithm; classically hard

- A_0, A_1 are known $\Rightarrow$ both **candidate signals** are computable, $f_b(t) = \langle v|e^{-iA_b t}|v\rangle$ for $b \in \{0,1\}$, and f is one of them
- find the time t^* at which they differ the most
- run the evolution at t^*, and see which value comes back
- the small graphs give both: the time t^*, and the two values $f_0(t^*)$ and $f_1(t^*)$

Quantum side: easy

- controlled e^{-iHt^*}, with a Hadamard test to estimate $f(t^*)$

Classical side: hard

- no generic classical way to evaluate a signal living in 2^N dimensions

The algorithm: the classical half

- **before the machine is on:** A_0 and A_1 are small, so this is ordinary linear algebra

$$t^* \;=\; \text{the time at which } f_0 \text{ and } f_1 \text{ differ the most}$$

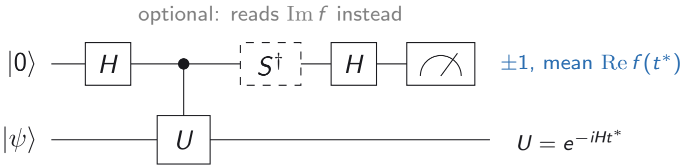
$$\mathrm{Dis} \;=\; \left| f_0(t^*) - f_1(t^*) \right|$$

- no sampling: t^* and Dis are fixed numbers, computed once

Choosing t^* costs no quantum computer at all.

The algorithm: the quantum half

- **at $t = t^*$:** a **Hadamard test**, one controlled e^{-iHt^*} per repetition
- $N_{\mathrm{rep}} = \mathcal{O}(1/\mathrm{Dis}^2)$ of them



Reading the signal needs a quantum interferometer.

Two instances

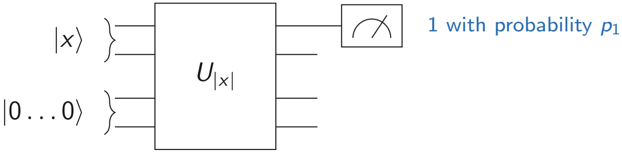
	big Hamiltonian H	small graph A_b
BQP	clock $\otimes$ circuit	C_m vs. C_{2m}
spired graphs	spired prism / Möbius G_{spire}	towered graph G_{tower}

- **BQP** — deciding whether C_m or C_{2m} hides inside the Hamiltonian captures the *full power of quantum computation*. A **theorem**.

- **spired graphs** — deciding whether a *prism* or a *Möbius ladder* hides inside the big obfuscated graph is **conjectured** to be classically hard, in the **black-box model**.

Formalizing the power of quantum computing

A language $L \subseteq \{0,1\}^*$: given an input x, decide whether $x \in L$.



- the circuit $U_{|x|}$ is built from $\mathrm{poly}(|x|)$ elementary gates, on N qubits: the input x, plus ancillas
- one **output qubit** is measured $\Rightarrow$ outcome 1 with probability p_1
- **promise:** $x \in L \Rightarrow p_1 > 2/3, \qquad x \notin L \Rightarrow p_1 < 1/3$

That is BQP. One bit: accept or reject.

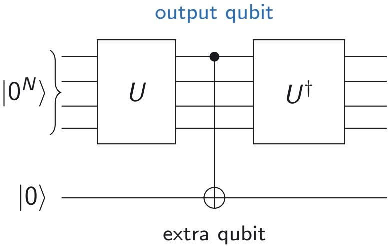
Two simplifications, both for free

- **Absorb the input.** Flip the qubits where x has a 1
  $\Rightarrow$ x is part of the circuit; the start is always $|0^N\rangle$
- **Amplify the gap.** Repeat and take the majority
  $\Rightarrow$ w.l.o.g. $p_1 = 0$ (**NO**) or $p_1 = 1$ (**YES**)

One circuit U, start $|0^N\rangle$, answer one bit b.

Step 1: from a circuit to an involution

- adjoin **one extra qubit**; the circuit copies its answer onto it, then undoes itself
- start from $|\mathrm{init}\rangle = |0^N\rangle|0\rangle$

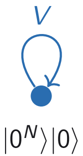


$$V \;=\; \big(U^\dagger \otimes I\big)\,\mathrm{CNOT}\,\big(U \otimes I\big)$$

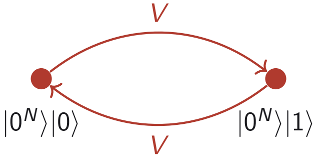
A Hermitian involution: $V = V^\dagger, \quad V^2 = I$

The answer is the dimension

- $V^2 = I \;\Rightarrow\;$ the orbit of $|\mathrm{init}\rangle$ is **one point or two**



NO ($p_1 = 0$): dim $= 1$



YES ($p_1 = 1$): dim $= 2$

- in the YES case the extra qubit flips, so V **swaps** two states instead of fixing one

Step 2: from the involution to a local Hamiltonian

- factor $V = V_m \cdots V_1$ into m elementary gates, add a clock register of dimension m
- define the **forward-time operator**:

$$
\begin{aligned}
F &= \sum_{t=0}^{m-1} |t{+}1 \bmod m\rangle\langle t|_c \otimes V_{t+1} \\
H &= F + F^\dagger
\end{aligned}
$$

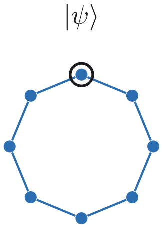
- F *walks* on the clock, a cycle of length m; each step applies one gate to the computational register
- H is $O(1)$-**local**, with $\mathrm{Spec}(H) \subseteq [-2, 2]$
- start from $|\psi\rangle = |0\rangle_c \otimes |\mathrm{init}\rangle$

H has dimension $m \cdot 2^{N+1}$.

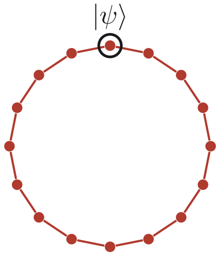
The two cycles

One lap of the clock applies the whole of V:

$$F^m|\psi\rangle = |0\rangle_c \otimes V|\mathrm{init}\rangle$$



NO: C_m
one lap comes back



YES: C_{2m}
two laps to come back

The time evolution is a walk on a cycle

- H acts on $m \cdot 2^{N+1}$ dimensions
- but from $|\psi\rangle$ the evolution never leaves the orbit of F:
  - m states in the **NO** case
  - $2m$ states in the **YES** case
- the orbit is orthonormal and H-invariant
- each vertex is the circuit after k gates: $|k\rangle_c \otimes V_k \cdots V_1 |\mathrm{init}\rangle$
- $\Rightarrow$ one step along the cycle is **one gate**

Walking the cycle is running the circuit.

The signal, winding by winding

- the walk on an **infinite line** is a Bessel function:

$$\langle x|e^{-iAt}|0\rangle \;=\; (-i)^x J_x(2t) \qquad \text{[Jacobi–Anger]}$$

- a cycle is that line **wrapped** $\Rightarrow$ sum over the images, one per **winding** $x = jn$:

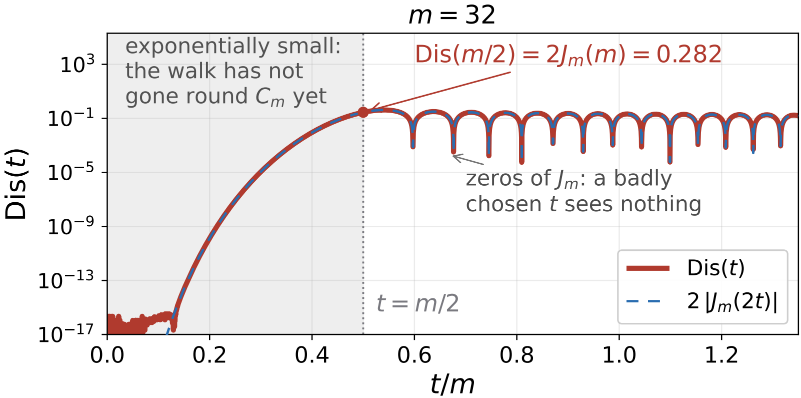
$$n = 2^b m, \quad b \in \{0,1\} \qquad\qquad f_b(t) \;=\; \sum_{j\in\mathbb{Z}} (-i)^{jn} J_{jn}(2t)$$

- the **even** windings are shared, and cancel:

$$f_0(t) - f_1(t) \;=\; \sum_{j \text{ odd}} (-i)^{jm} J_{jm}(2t) \;\approx\; 2\,(-i)^m J_m(2t)$$

One Bessel function, to an excellent approximation.

What the difference looks like



$\mathrm{Dis}(t) = \left|f_0(t) - f_1(t)\right|$ at $m = 32$, log scale. Dashed: the first odd winding, $2|J_m(2t)|$.

Exponentially small until the walk has gone once round C_m; then the first odd winding.

When to look, and how many times

- the amplitude at distance x is $J_x(2t)$, and J_x is exponentially small until its argument reaches its order
- $\Rightarrow$ **light cone**: distance $2t$ by time t
- at $t = m/2$: **once round** C_m, only **half way** round C_{2m}

$$t^* \;=\; \tfrac{m}{2}, \qquad \mathrm{Dis} \;=\; 2\,J_m(m) \;=\; \Theta\big(m^{-1/3}\big)$$

- $\Rightarrow$ $\mathcal{O}(m^{2/3})$ repetitions, polynomial

In BQP we did not choose the graph

- one natural reduction: a circuit $\Rightarrow$ a clock Hamiltonian $\Rightarrow$ a walk on C_m or C_{2m}

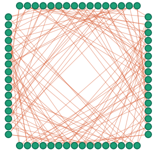
Now hide a graph we choose, inside a graph.

Dramatis personae

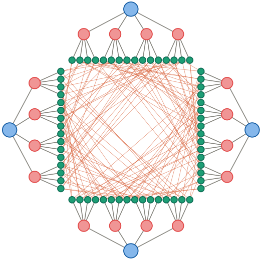
$$
\begin{tikzcd}
\overset{\text{lifted graph}}{G_{\text{lift}}} \arrow[rr, "\text{crown with spires}"] & & \overset{\text{spired graph}}{G_{\text{spire}}} \arrow[d, "\text{collapses to}"'] \\
\underset{\text{base graph}}{G} \arrow[u, "\text{replace each vertex}"'] & & \underset{\text{towered graph}}{G_{\text{tower}}}
\end{tikzcd}
$$

- pick *any* regular G on n vertices $\Rightarrow$ G_{spire} is complicated, and G **determines its symmetry**; the symmetry is what will collapse the walk

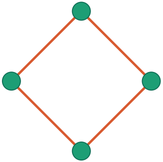
Dramatis personae



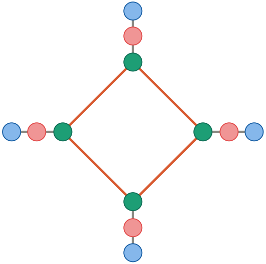
lifted graph



spired graph

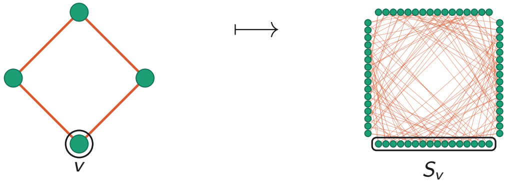


base graph



towered graph

The lift



base graph G

lifted graph

- every vertex becomes a **cluster** of look-alikes; every edge of G becomes a **bipartite graph** between the two clusters
- **regular** — the same number of partners for everyone $\Rightarrow$ the walk **never leaves** the uniform cluster states
- **random**, and every label **shuffled** $\Rightarrow$ every vertex looks alike: degree D, and no landmark anywhere

Regular buys the collapse. Random buys the hiding.

But now we are stuck too

- after the lift, *being at v* means the uniform state over v's cluster:

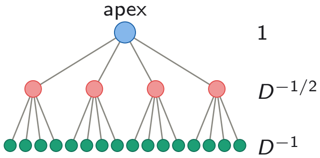
$$|S_v\rangle \;=\; |S_v|^{-1/2} \sum_{x \in S_v} |x\rangle$$

- the graph is reached only through an **oracle**, and every label is a **random string**; we cannot list a cluster's members
- $\Rightarrow$ it cannot hand us $|S_v\rangle$

To walk on G we need a state we cannot prepare.

The spire is the way in

- crown each cluster with a **tree** of height L: one **apex** on top
- the apex is a *single label*, which the oracle *can* hand over
- $\Rightarrow$ L steps down land exactly on $|S_v\rangle$



ℓ steps down: D^{ℓ} vertices, amplitude $D^{-\ell/2}$ each

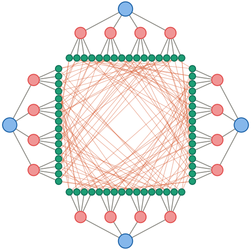
The spire turns the one label the oracle can give into the state we could not prepare.

Two collapses

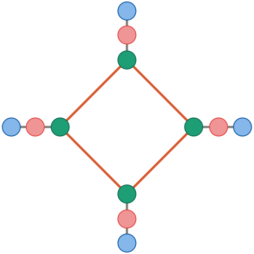
- the walk never leaves the span of the **level states**: every level, not just the cluster
- $\Rightarrow$ each **spire** becomes a **tower**: tree $\longrightarrow$ path
- $\Rightarrow$ each **cluster** becomes a **vertex**: D^L look-alikes $\longrightarrow$ one

G_{tower} **is the quotient of** G_{spire} **under the level-set partition.**

An exponentially large walk, run by an n^2 matrix



G_{spire}: $\Theta(nD^L)$ vertices, *hidden*



G_{tower}: $n(L+1) = n^2$ vertices, $L = n-1$, *on paper*

$$H|_{\mathcal{K}} = A_{\mathrm{tower}}$$

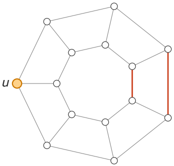
The obfuscating oracle

- **in**: a label $\Rightarrow$ **out**: the labels of its neighbours, and **nothing** else ($\perp$ if it is not a label); the **same** oracle serves the classical algorithm
- labels are random a-bit strings, $a = 2\log_2|V_{\text{spire}}| \Rightarrow$ a random string is a label with probability $\leq 1/|V_{\text{spire}}|$
- we are given **one** label: one apex, our entry point

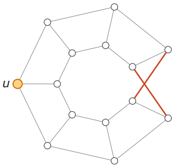
You cannot guess your way in. You can only walk.

Now we choose what to hide

All of that works for *any* regular G. So take the **hardest** pair.



prism Y_7



Möbius ladder M_7

- 3-regular on $n = 2m$ vertices: a ladder closed straight, or twisted
- two red edges in each: the *entire* difference
- u: our entry point, kept far from both

One oracle, two candidates

- **given**: H, through the oracle only; $\Theta(nD^L)$ vertices, nothing else
- **start**: the apex of one spire, a single label
- **promised**: on the subspace the walk explores, H acts as A_{tower}, built either from the **prism** or from the **Möbius ladder**
- $\Rightarrow$ decide which

Everything that remains is classical: two $n^2 \times n^2$ matrices, and where their signals differ most.

What the towers do to the spectrum

- we need both candidate signals, $f(t) = \sum_\lambda w_\lambda \, e^{-i\lambda t}$ over the spectrum of A_{tower}
- $n^2 \times n^2 \;\Rightarrow\; \mathcal{O}(n^6)$: hopeless at $n \approx 10^4$
- but the towers give A_{tower} a structure, and a symmetry:

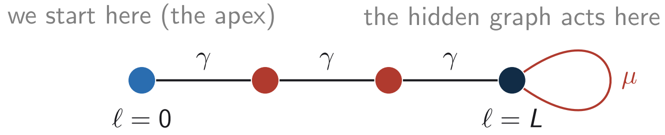
$$A_{\text{tower}} \;=\; A \otimes |L\rangle\langle L| \;+\; I \otimes P \qquad\qquad \big[\, A_{\text{tower}},\; A \otimes I \,\big] \;=\; 0$$

So A_{tower} splits into one block per eigenvalue of the hidden graph.

A direct sum of tridiagonal matrices

$$A_{\text{tower}} \;\cong\; \bigoplus_{j=1}^{n} T^{(\mu_j)}, \qquad T^{(\mu)} \;=\; \mu|L\rangle\langle L| + P$$

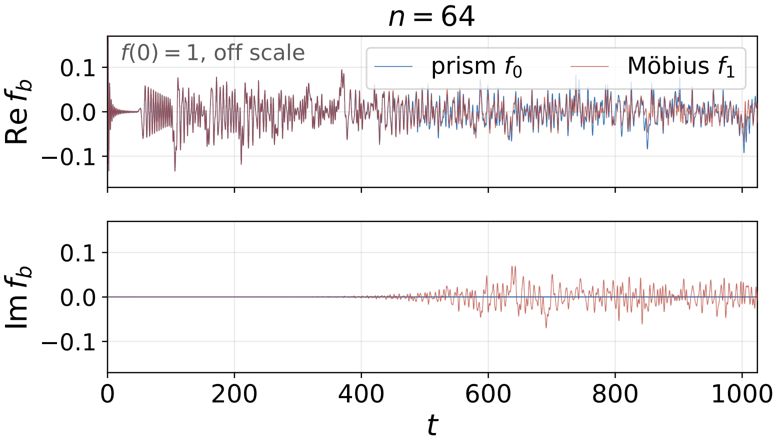
$\mu_1, \ldots, \mu_n$: the eigenvalues of the hidden graph. Each block is **tridiagonal**: the path with weight γ, and a single diagonal entry μ at the bottom.



- eigenvalues and eigenvectors from a **Chebyshev recurrence**
- n **independent** scalar problems $\Rightarrow$ the whole signal in $\widetilde{\mathcal{O}}(n^2)$

The boundary potential is the hidden graph's eigenvalue.

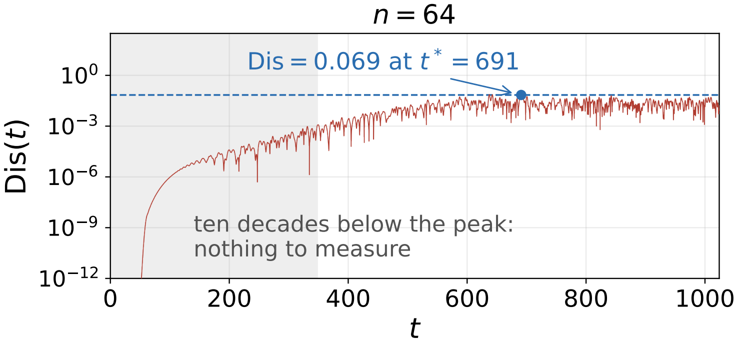
The two signals



Both signals decay to a band of size $\Theta(1/n)$ and stay there.

The real parts are one band; the imaginary parts are not.

Their difference



$\mathrm{Dis}(t)$, log scale: exponentially small until the walk reaches the two edges the candidates differ in, then a band whose peak is $\mathrm{Dis} = 0.069$ against a typical 0.027.

The peak stands $2.6\times$ above the typical level.

Why there is a peak, and what it costs

- $f_0 - f_1$ is about n^2 phases of size about $1/n^2 \;\Rightarrow\; \mathrm{Dis}$ is typically $1/n$ at a random time
- at t^* enough of them align to stand $\sqrt{\log n}$ *above* that level $\Rightarrow\; \mathrm{Dis} = \Theta\big(\sqrt{\log n}\,/\,n\big)$ [NUMERICS, $n \le 10^4$]
- and t^* is not just somewhere inside the horizon: it sits at $0.16\,n^2$ at every size tested

quantum	$\sim n^2$ repetitions, $\widetilde{\mathcal{O}}(n^4)$ gates	[CONJ]
classical	exponentially many queries	[CONJ]

The spires give us the cluster superposition, and with it the collapse; they are also why t^* is computed and not derived.

What is proved, and what is not

- **The mechanism is proved**
  - the walk really is C_m or C_{2m}
  - the collapse $H|_{\mathcal{K}} = A_{\mathrm{tower}}$
  - the spectral theory of G_{tower}
- **The separation is not**: two *conjectures*
  - classical hardness, *relative to the oracle*
  - the total quantum cost, $\widetilde{\mathcal{O}}(n^4)$ gates

Hidden subgroup, welded trees, and this one

- **hidden subgroup** — *which one is it?* answered with a group to exploit (discrete log, Shor)
- **welded trees** — *can you get there?* no group, and a proved lower bound
  - G_{spire} specializes to it at $G = K_2$
- **here** — *which one is it?* with no group to exploit

All of it is relative to the oracle.

Open:

- **exit-finding**: from the seed apex, find the *other* apices; the bridge between the two questions
- what else can play the **spire**'s part: funnel the amplitude in, look like nothing from outside
- find this structure arising *on its own*, somewhere in mathematics

Where this comes from

- **this talk** — *Quantum algorithm for identifying hidden graphs*, arXiv:2605.11228
- **BQP-complete problems** — Janzing & Wocjan, *Theory of Computing* 2007, *QIC* 2010
- **spectral graph theory** — with C. Elphick, 2013–2019
- **welded trees** — Childs et al., *STOC* 2003

The BQP construction is not part of the spired graph paper; this is the context it sits in.